

Dave's Daily

AI • Bayesian optimization • US equities

tl;dr

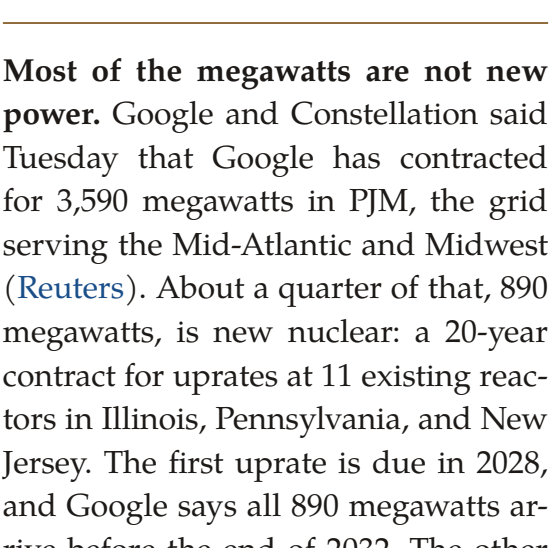
1. OpenAI published 722 math manuscripts in 372 families from an unreleased model. It posed about 4,000 problems and released reasoning summaries for only ten. Mathematicians disagree on whether the dump helps.
2. Google's Constellation contract is billed at 3,590 megawatts. Only 890 of those megawatts are new nuclear. The other 2,700 are a supply deal not tied to a plant.
3. A limit-order book filled only with reinforcement-learning agents either empties or freezes. Too few liquidity providers and the book collapses. Too many, packed into too little visible depth, and price discovery stops.

AI

A catalogue, not a model. On Tuesday evening OpenAI put 722 manuscripts, grouped into 372 result families, on GitHub under Apache 2.0 ([openai/math](#); [Scientific American](#)). The model is unreleased. The readme says the usual procedure posed about 4,000 open problems to one internal model and spent, on average, three hours of ChatGPT Pro thinking compute per result. Two items sat outside that procedure: a zero-free region for the Riemann zeta function, and the Hodge conjecture for CM abelian varieties. The write-up of the stronger zeta bound, real part above $11/12$, was edited by a person. Ten families come with an abridged reasoning summary (chart). OpenAI says some unformalized results could have issues, and that not every manuscript has a Lean proof.

People cannot check the method. An OpenAI spokesperson told Scientific American that almost every result came from one prompt to one agent, then added that some took multiple attempts, and that many of the new results are not yet understood by the company's own mathematicians. Andrew Sutherland of MIT says one-shot claims should stay unverified until the model can be replicated: "We should ask for receipts." Daniel Litt of Toronto says publishing the answers is good for mathematics. Scientific American reports that Terence Tao has called the pace of frontier labs' AI-generated results "insane." On 21 September OpenAI assembled an advisory group whose recommendations ask a lab to publish the model, the exact prompt, and the compute time. OpenAI published an average compute time and no prompts, and told Scientific American it is not bound by those recommendations.

What the math release actually contains



Markets

Most of the megawatts are not new power. Google and Constellation said Tuesday that Google has contracted for 3,590 megawatts in PJM, the grid serving the Mid-Atlantic and Midwest ([Reuters](#)). About a quarter of that, 890 megawatts, is new nuclear: a 20-year contract for uprates at 11 existing reactors in Illinois, Pennsylvania, and New Jersey. The first uprate is due in 2028, and Google says all 890 megawatts arrive before the end of 2032. The other 2,700 megawatts are a 15-year supply agreement not tied to a specific plant. Reuters says that piece is revenue certainty for plants already running. Hypothesis: a headline of 3.6 gigawatts overstates new supply by about three to one.

Who pays, and who gets the software. Constellation will invest more than \$4.3 billion. Google's own account says the uprates sustain about 4,400 jobs and support about 7,200 construction jobs, and that other grid customers bear none of the uprate cost ([Google](#)). Google also says Constellation will use Gemini Enterprise to plan the uprates and run the plants.

A book of agents can die in two ways. Jan Rosenzweig stocks a limit-order book with reinforcement-learning agents only ([arXiv:2609.31260](#), 25 Sep). Liquidity providers quote out to a chosen depth. Scalpers see only the touch and the next price. One agent hits the book with market orders. Too few liquidity providers and the book empties, with collapse probability above 90%. Too many of them in too little visible depth, and volatility falls below one tick, so price discovery freezes. Between those edges the book stays liquid. These phases are from the simulation, not from live US equities.

Key numbers (this edition)

Item	Figure
Problems posed, math release	≈4,000
Reasoning summaries released	10 of 372
New nuclear in the Google deal	890 MW
Supply not tied to a plant	2,700 MW
Evaluations in the BO study	110

Bayesian optimization

Scaling: search the function, not a fixed grid of knobs. Many "high-dimensional" problems are a curve or a shape that someone chopped into a vector before the optimizer ever ran. Davide Sartor, Donghyun Kim, Meghan Huber, and Nathan Wycoff (UMass Amherst) refuse that chop ([arXiv:2610.07417](#), 5 Oct; NeurIPS 2026). The thing being chosen is a function. The space of functions is infinite-dimensional.

The mechanism. They stay inside a reproducing kernel Hilbert space, but only on functions that are a sum of a few kernel bumps. Call the number of bumps m . Both the heights and the locations of the bumps are free, so the bumps can move. That set is a union of manifolds, nested as m grows: one bump, then two, and so on. A Gaussian process on those functions uses expected improvement. Multi-start L-BFGS-B maximizes it. Older functional methods lock the basis, or a random subspace, before the search starts, so they can only remix shapes the setup already imagined.

Data efficiency. The study uses the same budget for every method: 10 starting designs, then m (or the rival method's complexity) rises from 1 to 10, one step every 10 acquisitions. That is 110 evaluations in an infinite-dimensional domain. Restricting the search to sparse functions is how they keep the surrogate cheap enough to fit.

The evidence. They report the median over 16 to 32 runs, with a 95% interval for that median, on classical benchmarks lifted into function space and on control tasks (pendulum, hopper) plus a functional hyperparameter problem. LOMO, their method, had the best or tied-best median on most of them. The exception they flag is the brachistochrone, whose best shape is a smooth cycloid: a method that refines a smooth incumbent beat the sparse bumps there. There is no regret proof. The acquisition landscape is non-convex, so the optimizer can miss the best bumps. Code is at [bofus](#). Hypothesis: the gain, as in yesterday's one-lengthscale result, comes from asking the model to represent less. Here the limit is the number of bumps, not the number of lengthscales.